

# Exact and Delayed Descent in Accelerated Odd Collatz Spines with AAS-Based Metamorphic Separation

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## Abstract

This paper studies exact and delayed first-descent behavior in modular resistance spines for the accelerated odd Collatz map

$$T(n) = \frac{3n + 1}{2^{v_2(3n+1)}}$$

on positive odd integers. For spine elements

$$n = 2^m q - 1, \quad m \geq 2,$$

the persistence identity

$$T^j(n) = 3^j 2^{m-j} q - 1, \quad 0 \leq j \leq m - 1,$$

implies that no descent below the initial value can occur before step  $m$ . The paper isolates the valuation threshold

$$\kappa_m = \left\lceil \log_2 \left( \frac{3^m}{2^m - 1} \right) \right\rceil$$

and proves that the certified residue class

$$q \equiv q_m^* \pmod{2^{\kappa_m}}, \quad q_m^* \equiv (3^m)^{-1} \pmod{2^{\kappa_m}},$$

satisfies exact first descent:

$$\tau(2^m q - 1) = m.$$

Thus each spine contains an explicit infinite arithmetic subclass with descent at the first possible exit step.

The complementary non-certified classes are analyzed through shifted residues. If

$$q \equiv q_m^* + a \pmod{2^{\kappa_m}}, \quad 1 \leq a < 2^{\kappa_m},$$

then

$$v_2(3^m q - 1) = v_2(a).$$

This gives an infinite first-exit valuation structure for the non-certified complement, showing that failure of the certified threshold is algebraically controlled by the shift  $a$ .

Exact-integer computation up to  $N = 10^7$  was performed for all admissible spine levels  $2 \leq m \leq 23$ . The certified group contained 1,068,631 cases, all satisfying exact first descent at step  $m$ . The full non-certified complement contained 3,931,361 cases; all were resolved in the tested range, none descended at the first possible exit step, and all exhibited delayed descent, with

$$\overline{E} = 6.7360, \quad \text{median}(E) = 4, \quad \max E = 153,$$

where  $E(n) = \tau(n) - m$  denotes the pointwise delay beyond the deterministic lower bound. An AAS-based metamorphic post-evaluation separated the same regimes diagnostically: certified cases produced zero AAS, whereas all evaluated non-certified groups produced positive severity-sensitive scores.

The results do not prove the Collatz conjecture. Rather, they establish a local first-descent framework for the spine family  $n = 2^m q - 1$ : infinite certified exact-descent subclasses, an infinite first-exit valuation structure for shifted non-certified classes, and finite-range evidence for delayed descent throughout the tested full non-certified complement.

**Keywords:** Collatz conjecture; accelerated odd Collatz map; first-descent time; modular resistance spines; 2-adic valuation; exact descent; delayed descent; metamorphic testing; Artificial Age Score; finite exact verification.

## 1 Introduction

The Collatz problem concerns the long-term behavior of the iteration

$$C(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ 3n + 1, & n \equiv 1 \pmod{2}. \end{cases}$$

The conjecture asserts that every positive integer eventually reaches 1 under repeated iteration of  $C$ . Despite its elementary formulation, the problem remains open. Substantial literature exists on stopping times, congruence structures, and computational verification in the  $3x + 1$  problem [1, 2, 3]. A common reduction is obtained by restricting attention to positive odd integers and removing all powers of two after each odd step. This gives the accelerated odd Collatz map

$$T(n) = \frac{3n + 1}{2^{v_2(3n+1)}},$$

where  $v_2(x)$  denotes the exponent of the highest power of 2 dividing  $x$ . The map  $T$  sends positive odd integers to positive odd integers and records the odd-to-odd evolution of the Collatz process.

Rather than attempting to resolve global convergence, the present work focuses on local first-descent behavior. For a positive odd integer  $n > 1$ , the first-descent time is defined by

$$\tau(n) = \min\{r \geq 1 : T^r(n) < n\},$$

whenever such an  $r$  exists. This quantity measures the first time at which the accelerated odd trajectory falls below its initial value. A descent below the initial value is a local event, and the study of such events provides an intermediate structure between individual orbit computation and the full Collatz conjecture.

The present paper studies exact and delayed first descent inside modular resistance spines. For an integer

$$m \geq 2,$$

define the  $m$ -th spine by

$$S_m = \{n \in \mathbb{O}^+ : n \equiv -1 \pmod{2^m}\}.$$

Equivalently, every element of  $S_m$  has the form

$$n = 2^m q - 1, \quad q \geq 1.$$

A previously introduced pre-descent resistance framework established that such spine elements satisfy the persistence identity

$$T^j(n) = 3^j 2^{m-j} q - 1, \quad 0 \leq j \leq m - 1$$

[6]. Consequently,

$$T^j(n) > n, \quad 1 \leq j \leq m-1,$$

and therefore

$$\tau(n) \geq m$$

whenever  $\tau(n)$  exists. Thus membership in  $S_m$  forces a finite pre-descent delay: no descent below the initial value can occur before the  $m$ -th accelerated odd iterate.

The central question of this paper is whether the forced pre-descent delay

$$\tau(n) \geq m$$

inside a modular resistance spine can be converted into an explicit exact first-descent certificate for infinite subclasses, and whether the complementary non-certified residue classes admit an algebraic description at the first possible exit step.

The first part of this question asks when descent occurs exactly at the first possible exit step. For

$$n = 2^m q - 1,$$

the first possible exit value is governed by

$$T^m(n) = \frac{3^m q - 1}{2^{v_2(3^m q - 1)}}.$$

This motivates the valuation threshold

$$\kappa_m = \left\lceil \log_2 \left( \frac{3^m}{2^m - 1} \right) \right\rceil.$$

The exact-descent certificate proved below states that

$$v_2(3^m q - 1) \geq \kappa_m \implies \tau(2^m q - 1) = m.$$

Equivalently, since  $3^m$  is odd and therefore invertible modulo  $2^{\kappa_m}$ , the congruence

$$q \equiv (3^m)^{-1} \pmod{2^{\kappa_m}}$$

defines an infinite certified exact-descent subclass inside each spine.

This certified class is denoted by

$$\mathcal{D}_m = \{ 2^m q - 1 : q \equiv q_m^* \pmod{2^{\kappa_m}} \},$$

where

$$q_m^* \equiv (3^m)^{-1} \pmod{2^{\kappa_m}}.$$

Every element of  $\mathcal{D}_m$  satisfies

$$\tau(n) = m.$$

Thus the certified class gives exact first descent at the earliest step permitted by the spine persistence identity.

The complementary class is the full non-certified complement

$$\mathcal{N}_m = S_m \setminus \mathcal{D}_m.$$

Equivalently,

$$\mathcal{N}_m = \{ 2^m q - 1 : q \not\equiv q_m^* \pmod{2^{\kappa_m}} \}.$$

Elements of  $\mathcal{N}_m$  are not certified by the valuation threshold to descend at the first possible exit step. This does not imply non-descent. It only means that the exact first-exit certificate does not apply.

The second main algebraic result answers the first-exit part of the question for the non-certified complement. For each non-certified residue, write its unique shift from the certified residue as

$$a \in \{1, \dots, 2^{\kappa_m} - 1\}.$$

Then, for all shifted residue classes

$$q \equiv q_m^* + a \pmod{2^{\kappa_m}},$$

it is proved that

$$v_2(3^m q - 1) = v_2(a).$$

Thus the failure of the certified threshold in shifted non-certified classes is not arbitrary. It is exactly controlled by the 2-adic valuation of the shift from the certified residue. This gives an infinite algebraic description of the first-exit behavior of the non-certified complement.

The novelty of the present work should be understood in this local and structural sense. Stopping-time questions, accelerated odd dynamics, and congruence classes modulo powers of two are classical in the  $3x + 1$  literature. The contribution here is not the use of congruences or 2-adic valuations alone, but the isolation of an explicit finite valuation threshold inside the spine

$$n = 2^m q - 1$$

which converts pre-descent persistence into an exact first-descent certificate, together with a certified/non-certified partition

$$\mathcal{S}_m = \mathcal{D}_m \sqcup \mathcal{N}_m$$

and a shifted valuation description of the non-certified complement.

The residue structure also induces a natural metamorphic interpretation. The transformation

$$q \mapsto q + 2^{\kappa_m} t$$

preserves residue classes modulo  $2^{\kappa_m}$ . In the certified class, this transformation preserves the exact-descent certificate

$$\tau(2^m q - 1) = m.$$

In a shifted non-certified class, it preserves the first-exit valuation

$$v_2(3^m q - 1) = v_2(a).$$

Thus certified and shifted non-certified classes admit parallel metamorphic descriptions: one preserves exact first descent, while the other preserves structured failure of the first-exit threshold. This use of transformed input classes as consistency checks is related to the general philosophy of metamorphic testing [7, 8]. This perspective motivates the later AAS post-evaluation, where violations of the certified metamorphic relations are quantified after the exact algebraic classification has been established.

The algebraic results are complemented by exact-integer computation up to

$$N = 10^7.$$

Finite exact verification has played an important role in the computational study of the  $3x + 1$  problem [4]. In the present computation, all admissible spine levels

$$2 \leq m \leq 23$$

were tested, and the finite spine population was partitioned into the certified class

$$\mathcal{D}_m(10^7)$$

and the full non-certified complement

$$\mathcal{N}_m(10^7).$$

In this enumeration, a case means a spine-level instance  $(m, q)$ , equivalently an evaluated expression  $n = 2^m q - 1$  at a fixed spine level  $m$ , rather than a count of distinct odd starting values.

Across the tested range, the certified group contained

$$1,068,631$$

cases. All were resolved and all satisfied exact first descent at step  $m$ . The full non-certified complement contained

$$3,931,361$$

cases. Within this finite range, every tested non-certified case had a resolved first-descent time, none descended at the first possible exit step, and all exhibited delayed descent. The pointwise delayed-descent excess

$$E(n) = \tau(n) - m$$

satisfied

$$\overline{E} = 6.7360, \quad \text{median}(E) = 4, \quad \max E = 153$$

on the full non-certified complement.

An AAS-based metamorphic post-evaluation was also performed. The diagnostic comparison included the certified class, the full non-certified complement, and two non-certified control selections: boundary-shifted and random-spine cases. The certified group produced zero AAS in every tested case, whereas every evaluated case in the non-certified diagnostic groups produced a positive severity-sensitive score. The AAS framework used here is adapted from the Artificial Age Score framework [9]. In this paper, however, AAS is not used as a proof mechanism. It is used only as a diagnostic post-evaluation layer measuring finite-range separation between certified exact descent and non-certified delayed descent.

The paper makes three contributions. First, it proves an explicit valuation-threshold certificate for exact first descent inside modular resistance spines, producing infinite certified subclasses

$$\mathcal{D}_m$$

with

$$\tau(n) = m.$$

Second, it proves a shifted valuation lemma for the full non-certified complement

$$\mathcal{N}_m,$$

showing that first-exit threshold failure is exactly controlled by the 2-adic valuation of the shift from the certified residue. Third, it reports exact-integer finite-range evidence up to

$$N = 10^7$$

for evaluated spine-level cases, together with an AAS-based metamorphic post-evaluation separating certified exact descent from non-certified delayed descent.

These results do not prove the Collatz conjecture. They establish explicit infinite local exact-descent subclasses, prove an infinite first-exit valuation structure for shifted non-certified residue classes, and provide finite-range exact-integer evidence that the tested full non-certified complement exhibits delayed descent up to

$$N = 10^7.$$

## 2 Preliminaries and Notation

Let

$$\mathbb{O}^+ = \{n \in \mathbb{N} : n \equiv 1 \pmod{2}\}$$

denote the set of positive odd integers. The accelerated odd Collatz map is defined on  $\mathbb{O}^+$  by

$$T(n) = \frac{3n+1}{2^{v_2(3n+1)}}.$$

Here,  $v_2(x)$  denotes the largest exponent  $a \geq 0$  such that

$$2^a \mid x.$$

Since  $3n+1$  is even for every odd integer  $n$ , the quotient defining  $T(n)$  is again odd. Thus

$$T : \mathbb{O}^+ \rightarrow \mathbb{O}^+.$$

Iterates of  $T$  are denoted by

$$T^0(n) = n, \quad T^{r+1}(n) = T(T^r(n)).$$

For  $n \in \mathbb{O}^+$  with  $n > 1$ , the first-descent time is defined, whenever the following set is nonempty, by

$$\tau(n) = \min\{r \geq 1 : T^r(n) < n\}.$$

If no such  $r$  exists, then  $\tau(n)$  is left undefined.

For an integer  $m \geq 2$ , the  $m$ -th modular resistance spine is

$$S_m = \{n \in \mathbb{O}^+ : n \equiv -1 \pmod{2^m}\}.$$

Equivalently, every  $n \in S_m$  has a unique representation

$$n = 2^m q - 1, \quad q \geq 1.$$

The integer  $q$  is called the spine parameter.

For a spine element

$$n = 2^m q - 1,$$

the valuation governing the first possible exit from the spine is

$$s_m(q) = v_2(3^m q - 1).$$

The associated exact-descent threshold is

$$\kappa_m = \left\lceil \log_2 \left( \frac{3^m}{2^m - 1} \right) \right\rceil.$$

Equivalently,  $\kappa_m$  is the smallest integer  $k \geq 0$  such that

$$2^k(2^m - 1) \geq 3^m.$$

Since  $3^m$  is odd, it is invertible modulo  $2^{\kappa_m}$ . Define the certified residue

$$q_m^* \equiv (3^m)^{-1} \pmod{2^{\kappa_m}}.$$

The certified exact-descent class inside  $S_m$  is

$$\mathcal{D}_m = \{ 2^m q - 1 : q \geq 1, q \equiv q_m^* \pmod{2^{\kappa_m}} \}.$$

The full non-certified complement is

$$\mathcal{N}_m = S_m \setminus \mathcal{D}_m = \{ 2^m q - 1 : q \geq 1, q \not\equiv q_m^* \pmod{2^{\kappa_m}} \}.$$

Thus

$$S_m = \mathcal{D}_m \sqcup \mathcal{N}_m.$$

For

$$a \in \{0, 1, \dots, 2^{\kappa_m} - 1\},$$

define the shifted residue class

$$\mathcal{R}_{m,a} = \{ 2^m q - 1 : q \geq 1, q \equiv q_m^* + a \pmod{2^{\kappa_m}} \}.$$

Then

$$\mathcal{R}_{m,0} = \mathcal{D}_m,$$

and the full non-certified complement decomposes as the disjoint union

$$\mathcal{N}_m = \bigsqcup_{a=1}^{2^{\kappa_m}-1} \mathcal{R}_{m,a}.$$

Equivalently,

$$S_m = \bigsqcup_{a=0}^{2^{\kappa_m}-1} \mathcal{R}_{m,a}.$$

For

$$n = 2^m q - 1 \in S_m,$$

whenever  $\tau(n)$  exists, the delayed-descent excess relative to the deterministic spine lower bound is

$$E(n) = \tau(n) - m.$$

The case

$$E(n) = 0$$

corresponds to exact first descent at the first possible exit step, while

$$E(n) > 0$$

corresponds to delayed descent. If  $\tau(n)$  is undefined, then  $E(n)$  is also left undefined.

All inequalities involving iterates of  $T$  are taken in the ordinary order on positive integers. No global convergence statement is assumed.

### 3 Spine Persistence and Exit Formula

The following identity describes the deterministic behavior of the first  $m - 1$  accelerated odd iterates of an element in the  $m$ -th modular resistance spine. It is the algebraic mechanism that excludes early descent.

**Lemma 3.1** (Spine Persistence Identity). *Let*

$$m \geq 2, \quad q \geq 1,$$

*and define*

$$n = 2^m q - 1.$$

*Then, for every integer  $j$  satisfying*

$$0 \leq j \leq m - 1,$$

*one has*

$$T^j(n) = 3^j 2^{m-j} q - 1.$$

*In particular,*

$$T^j(n) > n, \quad 1 \leq j \leq m - 1.$$

*Proof.* The proof is by induction on  $j$ . For  $j = 0$ ,

$$T^0(n) = n = 2^m q - 1.$$

Assume that, for some  $j$  with

$$0 \leq j \leq m - 2,$$

one has

$$T^j(n) = 3^j 2^{m-j} q - 1.$$

Then

$$3T^j(n) + 1 = 3(3^j 2^{m-j} q - 1) + 1 = 3^{j+1} 2^{m-j} q - 2.$$

Hence

$$3T^j(n) + 1 = 2 \left( 3^{j+1} 2^{m-j-1} q - 1 \right).$$

Since  $j \leq m - 2$ , one has  $m - j - 1 \geq 1$ . Therefore

$$3^{j+1} 2^{m-j-1} q$$

is even, so

$$3^{j+1} 2^{m-j-1} q - 1$$

is odd. Thus

$$v_2(3T^j(n) + 1) = 1.$$

It follows that

$$T^{j+1}(n) = \frac{3T^j(n) + 1}{2} = 3^{j+1} 2^{m-j-1} q - 1.$$

This proves the induction step.

For  $1 \leq j \leq m - 1$ , the identity gives

$$T^j(n) > n$$

if and only if

$$3^j 2^{m-j} q - 1 > 2^m q - 1.$$



Equivalently,

$$3^j 2^{m-j} > 2^m.$$

After division by  $2^{m-j}$ , this becomes

$$3^j > 2^j,$$

which holds for every  $j \geq 1$ . Hence

$$T^j(n) > n, \quad 1 \leq j \leq m-1.$$

□

**Lemma 3.2** (Spine Exit Formula). *Let*

$$m \geq 2, \quad q \geq 1,$$

*and let*

$$n = 2^m q - 1.$$

*Define*

$$s = v_2(3^m q - 1).$$

*Then*

$$T^m(n) = \frac{3^m q - 1}{2^s}.$$

*Proof.* By Definition 3.1, with  $j = m-1$ ,

$$T^{m-1}(n) = 3^{m-1} 2q - 1.$$

Therefore

$$3T^{m-1}(n) + 1 = 3(3^{m-1} 2q - 1) + 1 = 2(3^m q - 1).$$

Since

$$s = v_2(3^m q - 1),$$

one has

$$v_2(3T^{m-1}(n) + 1) = s + 1.$$

Thus

$$T^m(n) = \frac{3T^{m-1}(n) + 1}{2^{s+1}} = \frac{2(3^m q - 1)}{2^{s+1}} = \frac{3^m q - 1}{2^s}.$$

□

**Corollary 3.3** (Pre-Descent Lower Bound). *Let*

$$m \geq 2, \quad q \geq 1,$$

*and let*

$$n = 2^m q - 1.$$

*If  $\tau(n)$  exists, then*

$$\tau(n) \geq m.$$

*Proof.* By Definition 3.1,

$$T^j(n) > n$$

for every

$$1 \leq j \leq m-1.$$

Therefore no descent below  $n$  can occur during the first  $m-1$  accelerated odd iterates. Hence, if a first-descent time exists, it must satisfy

$$\tau(n) \geq m.$$

□

## 4 Valuation Threshold for Exact First Descent

The persistence identity excludes descent before the  $m$ -th accelerated odd iterate for every element of the form

$$n = 2^m q - 1.$$

Thus exact first descent at time  $m$  occurs whenever the exit value

$$T^m(n)$$

falls below the initial value  $n$ . The following theorem gives a uniform valuation threshold that guarantees this event.

**Theorem 4.1** (Valuation Threshold for Exact First Descent). *Let*

$$m \geq 2, \quad q \geq 1,$$

*and define*

$$n = 2^m q - 1.$$

*Let*

$$s = v_2(3^m q - 1)$$

*and*

$$\kappa_m = \left\lceil \log_2 \left( \frac{3^m}{2^m - 1} \right) \right\rceil.$$

*If*

$$s \geq \kappa_m,$$

*then*

$$\tau(n) = m.$$

*Equivalently,*

$$v_2(3^m q - 1) \geq \kappa_m \implies \tau(2^m q - 1) = m.$$

*Proof.* By Definition 3.3, no descent below  $n$  can occur before step  $m$ . Hence the first-descent time, if it occurs, cannot be smaller than  $m$ .

It remains to show that descent occurs at step  $m$ . By Definition 3.2,

$$T^m(n) = \frac{3^m q - 1}{2^s}.$$

Therefore it is enough to prove

$$\frac{3^m q - 1}{2^s} < 2^m q - 1.$$

Multiplying by  $2^s$ , this is equivalent to

$$3^m q - 1 < 2^s (2^m q - 1).$$

Expanding the right-hand side gives

$$3^m q - 1 < 2^{m+s} q - 2^s.$$

After rearrangement, this becomes

$$q(2^{m+s} - 3^m) > 2^s - 1.$$

Since

$$s \geq \kappa_m$$

and  $\kappa_m$  is the smallest integer  $k \geq 0$  satisfying

$$2^k(2^m - 1) \geq 3^m,$$

one has

$$2^s(2^m - 1) \geq 3^m.$$

Equivalently,

$$2^{m+s} - 2^s \geq 3^m.$$

Thus

$$2^{m+s} - 3^m \geq 2^s.$$

Since  $q \geq 1$ ,

$$q(2^{m+s} - 3^m) \geq 2^{m+s} - 3^m \geq 2^s.$$

Hence

$$q(2^{m+s} - 3^m) > 2^s - 1.$$

Therefore

$$T^m(n) < n.$$

Since descent is impossible during the first  $m - 1$  accelerated odd iterates and occurs at step  $m$ , it follows that

$$\tau(n) = m.$$

□

**Proposition 4.2** (Congruence Form of the Threshold). *Let*

$$m \geq 2.$$

*Then*

$$v_2(3^m q - 1) \geq \kappa_m$$

*if and only if*

$$3^m q \equiv 1 \pmod{2^{\kappa_m}}.$$

*Equivalently,*

$$q \equiv (3^m)^{-1} \pmod{2^{\kappa_m}}.$$

*Proof.* The inequality

$$v_2(3^m q - 1) \geq \kappa_m$$

is equivalent to the divisibility condition

$$2^{\kappa_m} \mid 3^m q - 1.$$

This is equivalent to

$$3^m q \equiv 1 \pmod{2^{\kappa_m}}.$$

Since  $3^m$  is odd, it is invertible modulo  $2^{\kappa_m}$ . Therefore the congruence is equivalent to

$$q \equiv (3^m)^{-1} \pmod{2^{\kappa_m}}.$$

□

**Corollary 4.3** (Infinite Certified Exact-Descent Classes). *For every*

$$m \geq 2,$$

*the class*

$$\mathcal{D}_m = \{ 2^m q - 1 : q \equiv q_m^* \pmod{2^{\kappa_m}} \}$$

*is infinite, and every element*

$$n \in \mathcal{D}_m$$

*satisfies*

$$\tau(n) = m.$$

*Proof.* If

$$n \in \mathcal{D}_m,$$

then

$$n = 2^m q - 1$$

with

$$q \equiv q_m^* \pmod{2^{\kappa_m}},$$

where

$$q_m^* \equiv (3^m)^{-1} \pmod{2^{\kappa_m}}.$$

By Definition 4.2,

$$v_2(3^m q - 1) \geq \kappa_m.$$

Therefore Definition 4.1 gives

$$\tau(n) = m.$$

The class is infinite because it contains all values

$$n_t = 2^m(q_m^* + 2^{\kappa_m} t) - 1, \quad t \geq 0.$$

□

*Remark 4.4* (Certified and Non-Certified Regimes). The threshold condition

$$v_2(3^m q - 1) \geq \kappa_m$$

is a sufficient uniform condition for exact first descent at step  $m$ . Its failure does not imply non-descent; it only means that the first-exit descent certificate does not apply. The complementary non-certified classes are analyzed below through their shifted residue structure.

*Remark 4.5* (Local Scope). The exact-descent theorem is local to the spine

$$n = 2^m q - 1.$$

It does not assert global convergence of arbitrary Collatz trajectories and does not prove that every odd integer satisfies such a valuation condition.

## 5 Modular Exact-Descent Subclasses and Shifted Complements

The valuation threshold in Definition 4.1 defines one certified residue class inside each modular resistance spine. This section describes the corresponding exact-descent subclass, its full non-certified complement, and the shifted residue structure of the non-certified regime. By Definition 4.2, the valuation threshold is equivalent to the certified congruence

$$q \equiv (3^m)^{-1} \pmod{2^{\kappa_m}}.$$

**Definition 5.1** (Certified Exact-Descent Class). For each  $m \geq 2$ , let  $q_m^*$  denote the representative in

$$\{1, \dots, 2^{\kappa_m} - 1\}$$

satisfying

$$q_m^* \equiv (3^m)^{-1} \pmod{2^{\kappa_m}}.$$

The certified exact-descent class inside  $S_m$  is

$$\mathcal{D}_m = \{2^m q - 1 : q \geq 1, q \equiv q_m^* \pmod{2^{\kappa_m}}\}.$$

Equivalently,

$$\mathcal{D}_m = \{2^m(q_m^* + 2^{\kappa_m}t) - 1 : t \geq 0\}.$$

The infinitude and exact-descent property of  $\mathcal{D}_m$  follow from Definition 4.3.

**Definition 5.2** (Full Non-Certified Complement). The full non-certified complement of  $\mathcal{D}_m$  inside  $S_m$  is

$$\mathcal{N}_m = S_m \setminus \mathcal{D}_m.$$

Equivalently,

$$\mathcal{N}_m = \{2^m q - 1 : q \geq 1, q \not\equiv q_m^* \pmod{2^{\kappa_m}}\}.$$

Thus

$$S_m = \mathcal{D}_m \sqcup \mathcal{N}_m.$$

The non-certified complement consists of all residue classes modulo  $2^{\kappa_m}$  except the certified one. These classes are naturally indexed by shifts from the certified residue.

**Definition 5.3** (Shifted Residue Classes). For

$$a \in \{0, 1, \dots, 2^{\kappa_m} - 1\},$$

define

$$\mathcal{R}_{m,a} = \{2^m q - 1 : q \geq 1, q \equiv q_m^* + a \pmod{2^{\kappa_m}}\}.$$

Then

$$\mathcal{R}_{m,0} = \mathcal{D}_m,$$

and

$$\mathcal{N}_m = \bigsqcup_{a=1}^{2^{\kappa_m}-1} \mathcal{R}_{m,a}.$$

Equivalently,

$$S_m = \bigsqcup_{a=0}^{2^{\kappa_m}-1} \mathcal{R}_{m,a}.$$

The next lemma gives the exact first-exit valuation on each shifted non-certified class. It is the main algebraic structure behind the non-certified complement.

**Lemma 5.4** (Shift Valuation Lemma). *Let  $m \geq 2$ , and let*

$$q_m^* \equiv (3^m)^{-1} \pmod{2^{\kappa_m}}.$$

*Suppose that*

$$q \equiv q_m^* + a \pmod{2^{\kappa_m}}, \quad 1 \leq a < 2^{\kappa_m}.$$

*Then*

$$v_2(3^m q - 1) = v_2(a).$$

*Proof.* Since

$$q \equiv q_m^* + a \pmod{2^{\kappa_m}},$$

there exists an integer  $t \in \mathbb{Z}$  such that

$$q = q_m^* + a + 2^{\kappa_m} t.$$

Therefore

$$3^m q - 1 = (3^m q_m^* - 1) + 3^m a + 3^m 2^{\kappa_m} t.$$

By the definition of  $q_m^*$ ,

$$3^m q_m^* \equiv 1 \pmod{2^{\kappa_m}},$$

so

$$2^{\kappa_m} \mid 3^m q_m^* - 1.$$

Also,

$$2^{\kappa_m} \mid 3^m 2^{\kappa_m} t.$$

Hence

$$3^m q - 1 = 3^m a + 2^{\kappa_m} L$$

for some integer  $L$ .

Let

$$r = v_2(a).$$

Since

$$1 \leq a < 2^{\kappa_m},$$

one has

$$r < \kappa_m.$$

Because  $3^m$  is odd,

$$v_2(3^m a) = v_2(a) = r.$$

Using

$$3^m q - 1 = 3^m a + 2^{\kappa_m} L,$$

we obtain

$$3^m q - 1 = 2^r \left( 3^m \frac{a}{2^r} + 2^{\kappa_m - r} L \right).$$

The integer

$$3^m \frac{a}{2^r}$$

is odd, while

$$2^{\kappa_m - r} L$$

is even. Therefore the factor in parentheses is odd. Hence

$$v_2(3^m q - 1) = r = v_2(a).$$

□

**Corollary 5.5** (Structured Failure of the Certified Threshold). *Let*

$$1 \leq a < 2^{\kappa_m}.$$

*For every*

$$n = 2^m q - 1 \in \mathcal{R}_{m,a},$$

*the first-exit valuation is*

$$v_2(3^m q - 1) = v_2(a).$$

*In particular,*

$$v_2(3^m q - 1) < \kappa_m.$$

*Hence no element of  $\mathcal{R}_{m,a}$  satisfies the certified threshold*

$$v_2(3^m q - 1) \geq \kappa_m.$$

*Proof.* The equality

$$v_2(3^m q - 1) = v_2(a)$$

is exactly Definition 5.4. Since

$$1 \leq a < 2^{\kappa_m},$$

one has

$$v_2(a) < \kappa_m.$$

Therefore

$$v_2(3^m q - 1) < \kappa_m.$$

Thus the certified threshold cannot hold on any shifted class with

$$1 \leq a < 2^{\kappa_m}.$$

□

**Remark 5.6** (Interpretation of the Partition). The decomposition

$$\mathcal{S}_m = \mathcal{D}_m \sqcup \mathcal{N}_m$$

separates the spine into a certified exact-descent regime and a non-certified regime. On  $\mathcal{D}_m$ , the valuation threshold proves

$$\tau(n) = m.$$

On  $\mathcal{N}_m$ , the threshold fails at the first possible exit step, but this failure is algebraically structured by the shift parameter  $a$ . This does not imply non-descent. It only shows that the uniform first-exit threshold fails, so exact descent at step  $m$  is not obtained by the certified valuation mechanism.

## 6 Size and Interpretation of the Threshold

The threshold

$$\kappa_m = \left\lceil \log_2 \left( \frac{3^m}{2^m - 1} \right) \right\rceil$$

measures the amount of 2-adic divisibility required at the first possible exit step from the spine. By Definition 3.2,

$$T^m(2^m q - 1) = \frac{3^m q - 1}{2^{v_2(3^m q - 1)}}.$$

Thus descent at the first possible exit step is forced when the denominator contains enough powers of 2 to overcome the growth of the numerator relative to the initial value

$$2^m q - 1.$$

The threshold is adapted to the exact endpoint inequality

$$\frac{3^m q - 1}{2^s} < 2^m q - 1,$$

rather than to an asymptotic comparison alone. The defining inequality

$$2^{\kappa_m}(2^m - 1) \geq 3^m$$

is precisely the finite correction needed to account for the  $-1$  terms in

$$2^m q - 1$$

and

$$3^m q - 1.$$

This is why the denominator in the threshold is  $2^m - 1$ , not merely  $2^m$ .

**Proposition 6.1** (Asymptotic Size of the Threshold). *The threshold satisfies*

$$\kappa_m = \left\lceil m \log_2 \left( \frac{3}{2} \right) - \log_2(1 - 2^{-m}) \right\rceil.$$

*In particular,*

$$\kappa_m = m \log_2 \left( \frac{3}{2} \right) + O(1)$$

*as  $m \rightarrow \infty$ .*

*Proof.* Using

$$2^m - 1 = 2^m(1 - 2^{-m}),$$

one obtains

$$\frac{3^m}{2^m - 1} = \left( \frac{3}{2} \right)^m \frac{1}{1 - 2^{-m}}.$$

Taking logarithms gives

$$\log_2 \left( \frac{3^m}{2^m - 1} \right) = m \log_2 \left( \frac{3}{2} \right) - \log_2(1 - 2^{-m}).$$



Therefore

$$\kappa_m = \left\lceil m \log_2 \left( \frac{3}{2} \right) - \log_2(1 - 2^{-m}) \right\rceil.$$

Since

$$-\log_2(1 - 2^{-m}) \rightarrow 0$$

as  $m \rightarrow \infty$ , and the ceiling changes a real number by less than 1, the asserted asymptotic estimate follows.  $\square$

The modulus

$$2^{\kappa_m}$$

determines the spacing between certified spine parameters. Since the certified class selects one residue class modulo  $2^{\kappa_m}$ , its relative density among the spine parameters  $q$  is

$$2^{-\kappa_m}.$$

Using

$$\kappa_m = m \log_2 \left( \frac{3}{2} \right) + O(1),$$

this density has the asymptotic scale

$$2^{-\kappa_m} \asymp \left( \frac{2}{3} \right)^m.$$

Thus certified exact-descent classes become increasingly sparse as  $m$  grows.

The full non-certified complement has complementary relative density

$$1 - 2^{-\kappa_m}$$

inside the  $m$ -th spine. Hence most spine parameters are not certified by the first-exit threshold. This does not indicate non-descent; it only shows that most parameters require analysis beyond the first possible exit step.

The shifted residue decomposition developed above makes this distinction explicit. The certified class is

$$\mathcal{R}_{m,0} = \mathcal{D}_m,$$

while the full non-certified complement is the disjoint union

$$\mathcal{N}_m = \bigsqcup_{a=1}^{2^{\kappa_m}-1} \mathcal{R}_{m,a}.$$

On  $\mathcal{D}_m$ , the valuation threshold certifies

$$\tau(n) = m.$$

On each shifted non-certified class  $\mathcal{R}_{m,a}$ , the first-exit valuation is instead fixed by

$$v_2(3^m q - 1) = v_2(a).$$

Thus the threshold separates one exact first-descent residue class from a structured family of non-certified residue classes.

In this sense,  $\kappa_m$  has two simultaneous roles. First, it is the exact finite valuation threshold needed to force first-exit descent. Second, it determines the modular resolution at which the spine decomposes into certified and shifted non-certified regimes.

## Interpretation

The certified mechanism has two components. Spine membership gives the forced lower bound

$$\tau(n) \geq m,$$

while the valuation threshold

$$v_2(3^m q - 1) \geq \kappa_m$$

forces descent at the first possible exit step. Hence, on the certified class

$$q \equiv q_m^* \pmod{2^{\kappa_m}},$$

one obtains

$$\tau(2^m q - 1) = m.$$

The role of  $\kappa_m$  is therefore not only quantitative but structural. Quantitatively, it measures the number of powers of 2 required to overcome the growth from  $3^m$  at the exit step. Structurally, it determines the modulus

$$2^{\kappa_m}$$

at which the spine separates into the certified class

$$\mathcal{D}_m$$

and the full non-certified complement

$$\mathcal{N}_m.$$

The non-certified complement is not a failure set for descent. It is only the set where the first-exit certificate does not apply. Its shifted residue classes satisfy

$$q \equiv q_m^* + a \pmod{2^{\kappa_m}}, \quad 1 \leq a < 2^{\kappa_m},$$

and have first-exit valuation

$$v_2(3^m q - 1) = v_2(a).$$

Thus the certified class gives exact descent at the first possible exit step, while the non-certified classes provide the structured setting in which delayed descent is analyzed.

## 7 Exact-Integer Verification and AAS-Based Metamorphic Post-Evaluation

The algebraic results proved above do not depend on computation. Exact-integer verification was nevertheless performed for three purposes: to confirm the certified exact-descent classes over a finite range, to examine the full non-certified complement, and to compare certified classes with nearby and generic non-certified controls. This computational component is in the tradition of finite exact verification in the  $3x + 1$  literature, but it is not used as proof of the algebraic theorems [4].

All computations were performed using exact integer arithmetic. In particular, no floating-point approximations were used to evaluate

$$v_2(3^m q - 1),$$

the threshold

$$\kappa_m,$$

or the accelerated odd iterates. The threshold was computed using the integer condition

$$2^{\kappa_m}(2^m - 1) \geq 3^m,$$

with  $\kappa_m$  chosen minimally.

The search range was

$$N = 10^7.$$

Since

$$2^{23} - 1 = 8,388,607 < 10^7$$

and

$$2^{24} - 1 = 16,777,215 > 10^7,$$

the admissible spine levels in this range are precisely

$$2 \leq m \leq 23.$$

For each admissible  $m$ , the certified residue

$$q_m^* \equiv (3^m)^{-1} \pmod{2^{\kappa_m}}$$

was computed. Every spine-level case

$$n = 2^m q - 1 \leq 10^7$$

was then classified according to whether

$$q \equiv q_m^* \pmod{2^{\kappa_m}}$$

or

$$q \not\equiv q_m^* \pmod{2^{\kappa_m}}.$$

Here a case means an evaluated pair  $(m, q)$ , equivalently an instance  $n = 2^m q - 1$  at a fixed spine level  $m$ , rather than a count of distinct odd starting values.

Thus, for each  $m$ , the finite tested spine population was partitioned as

$$S_m(10^7) = \mathcal{D}_m(10^7) \sqcup \mathcal{N}_m(10^7),$$

where

$$\mathcal{D}_m(10^7) = \{n \in \mathcal{D}_m : n \leq 10^7\}$$

and

$$\mathcal{N}_m(10^7) = \{n \in \mathcal{N}_m : n \leq 10^7\}.$$

For every tested element, the accelerated odd trajectory was iterated until either a first descent was found or the prescribed iteration limit

$$L = 10000$$

was reached. The computed first-descent time was

$$\tau(n) = \min\{r \geq 1 : T^r(n) < n\},$$

when resolved. For spine-level cases, the delayed-descent excess was recorded as

$$E(n) = \tau(n) - m.$$

The algebraic lower bound

$$\tau(n) \geq m$$

was used only as a consistency check; the computed value of  $\tau(n)$  was obtained directly by exact iteration.

The computation was organized into two principal groups and two auxiliary diagnostic controls. The principal groups were the certified class

$$\mathcal{D}_m(10^7)$$

and the full non-certified complement

$$\mathcal{N}_m(10^7).$$

Together these form the non-overlapping partition of the tested spine population. The auxiliary groups were boundary-shifted cases and random-spine cases. These were not additional components of the partition; they were included only as diagnostic controls for the AAS-based post-evaluation.

The boundary-shifted group consisted of parameters in the residue class

$$q \equiv q_m^* + 1 \pmod{2^{\kappa_m}},$$

whenever such parameters occurred in the search range. This class is the nearest shifted non-certified class to the certified residue and satisfies

$$v_2(3^m q - 1) = v_2(1) = 0$$

by the shift valuation lemma. The random-spine group consisted of sampled non-certified spine parameters and was used as a generic non-certified diagnostic control.

For each tested case, an AAS-style post-evaluation score was computed from finite diagnostic discrepancies. The AAS layer was applied after the exact integer computation and was not used to determine whether descent occurred. Its role was to measure finite-range diagnostic separation between the certified regime and non-certified controls. In particular, certified cases were expected to satisfy the algebraic certificate exactly and therefore produce zero diagnostic score, whereas non-certified controls were expected to exhibit positive severity-sensitive scores corresponding to failure of the first-exit certificate and delayed descent.

The AAS computation used the following diagnostic quantities:

$$\Delta_{\text{residue}}, \quad \Delta_{\text{valuation}}, \quad \Delta_{\text{predescent}}, \quad \Delta_{\text{exit}}, \quad \Delta_{\tau}.$$

These quantities measured, respectively, residue-class agreement with the certified condition, agreement with the certified valuation threshold, consistency with the pre-descent lower bound, first-exit descent behavior, and agreement with exact descent at time  $m$ . The resulting AAS value was used only as a severity-sensitive diagnostic summary.

Thus the computational part has two distinct roles. The first is exact verification of the finite partition

$$S_m(10^7) = \mathcal{D}_m(10^7) \sqcup \mathcal{N}_m(10^7)$$

and the corresponding first-descent behavior. The second is post-evaluation of the certified/non-certified separation through AAS-based metamorphic diagnostics. Neither role replaces the algebraic proofs, and neither is used to assert global convergence.

## 7.1 Exact-Integer Verification of Certified Subclasses

For each certified case, exact integer iteration was used to verify the computed first-descent time against the algebraic prediction

$$\tau(n) = m.$$

Equivalently, the computation checked that no descent occurred before the  $m$ -th accelerated odd iterate and that

$$T^m(n) < n.$$

Across all admissible levels

$$2 \leq m \leq 23,$$

the finite certified population contained

$$1,068,631$$

cases. Every certified case was resolved and every one satisfied exact first descent at step  $m$ . No certified failure was observed.

| Quantity                        | Value              |
|---------------------------------|--------------------|
| Search bound $N$                | $10^7$             |
| Admissible $m$ -range           | $2 \leq m \leq 23$ |
| Certified cases tested          | 1,068,631          |
| Certified exact descents at $m$ | 1,068,631          |
| Certified failures              | 0                  |

Table 1: Exact-integer verification summary for certified subclasses up to  $N = 10^7$ .

Thus the finite verification agrees with the algebraic certificate of Definition 4.1 throughout the tested range.

The distribution of certified and non-certified cases by spine level is shown in Table 2. For fixed  $m$ , the number of tested spine elements is

$$|S_m(10^7)| = \left\lfloor \frac{10^7 + 1}{2^m} \right\rfloor.$$

The certified class selects one residue class modulo

$$2^{\kappa_m},$$

while the remaining residue classes form the full non-certified complement.

For

$$m \geq 16,$$

no certified element occurred below

$$N = 10^7.$$

This is a finite endpoint effect and does not contradict the infinitude of

$$\mathcal{D}_m$$

for every fixed

$$m \geq 2.$$

Indeed,  $\mathcal{D}_m$  remains infinite by the arithmetic progression

$$2^m(q_m^* + 2^{\kappa_m}t) - 1, \quad t \geq 0.$$

| $m$ | $\kappa_m$ | $2^{\kappa_m}$ | $ S_m(10^7) $ | $ \mathcal{D}_m(10^7) $ | $ \mathcal{N}_m(10^7) $ |
|-----|------------|----------------|---------------|-------------------------|-------------------------|
| 2   | 2          | 4              | 2,500,000     | 625,000                 | 1,875,000               |
| 3   | 2          | 4              | 1,250,000     | 312,500                 | 937,500                 |
| 4   | 3          | 8              | 625,000       | 78,125                  | 546,875                 |
| 5   | 3          | 8              | 312,500       | 39,063                  | 273,437                 |
| 6   | 4          | 16             | 156,250       | 9,766                   | 146,484                 |
| 7   | 5          | 32             | 78,125        | 2,442                   | 75,683                  |
| 8   | 5          | 32             | 39,062        | 1,221                   | 37,841                  |
| 9   | 6          | 64             | 19,531        | 306                     | 19,225                  |
| 10  | 6          | 64             | 9,765         | 153                     | 9,612                   |
| 11  | 7          | 128            | 4,882         | 38                      | 4,844                   |
| 12  | 8          | 256            | 2,441         | 10                      | 2,431                   |
| 13  | 8          | 256            | 1,220         | 5                       | 1,215                   |
| 14  | 9          | 512            | 610           | 1                       | 609                     |
| 15  | 9          | 512            | 305           | 1                       | 304                     |
| 16  | 10         | 1024           | 152           | 0                       | 152                     |
| 17  | 10         | 1024           | 76            | 0                       | 76                      |
| 18  | 11         | 2048           | 38            | 0                       | 38                      |
| 19  | 12         | 4096           | 19            | 0                       | 19                      |
| 20  | 12         | 4096           | 9             | 0                       | 9                       |
| 21  | 13         | 8192           | 4             | 0                       | 4                       |
| 22  | 13         | 8192           | 2             | 0                       | 2                       |
| 23  | 14         | 16384          | 1             | 0                       | 1                       |

Table 2: Certified and full non-certified spine counts up to  $N = 10^7$ . The threshold is  $\kappa_m = \lceil \log_2(3^m/(2^m - 1)) \rceil$ .

## 7.2 Delayed Descent in the Full Non-Certified Complement

The full non-certified complement consists of the tested spine elements outside the certified residue class. These elements do not satisfy the valuation-threshold certificate at the first possible exit step. Therefore, exact descent at time  $m$  is not certified algebraically for this group.

For each tested non-certified element, the first-descent time  $\tau(n)$  was computed by exact integer iteration. Whenever  $\tau(n)$  was resolved, the delayed-descent excess was recorded as

$$E(n) = \tau(n) - m.$$

The value

$$E(n) = 0$$

corresponds to exact descent at the first possible exit step, whereas

$$E(n) > 0$$

corresponds to delayed descent.

Across the finite range

$$N = 10^7,$$

the full non-certified complement contained

$$3,931,361$$

tested cases. All of these cases had resolved first-descent times within the iteration limit  $L = 10000$ . None descended at the first possible exit step  $m$ , and all exhibited delayed descent. The observed delayed-descent statistics were

$$\overline{E} = 6.7360, \quad \text{median}(E) = 4, \quad \max E = 153.$$

| Quantity                        | Value     |
|---------------------------------|-----------|
| Full non-certified cases tested | 3,931,361 |
| Resolved first-descent times    | 3,931,361 |
| Unresolved cases                | 0         |
| Exact descents at $m$           | 0         |
| Delayed descents                | 3,931,361 |
| Mean $\tau$                     | 9.8429    |
| Median $\tau$                   | 7         |
| Maximum $\tau$                  | 155       |
| Mean $E$                        | 6.7360    |
| Median $E$                      | 4         |
| Maximum $E$                     | 153       |

Table 3: Delayed-descent summary for the full non-certified complement up to  $N = 10^7$ .

The result should be interpreted as finite-range exact evidence, not as an infinite theorem. It shows that, up to

$$N = 10^7,$$

the full non-certified complement was resolved by exact integer computation and exhibited delayed descent rather than first-exit descent. It does not prove that all non-certified spine elements descend.

### 7.3 Metamorphic Structure of Certified and Shifted Classes

The residue decomposition modulo

$$2^{\kappa_m}$$

induces a natural metamorphic structure on the spine parameter  $q$ . The basic transformation is

$$q \mapsto q + 2^{\kappa_m} t, \quad t \geq 0.$$

This transformation preserves the residue class of  $q$  modulo  $2^{\kappa_m}$ , and therefore preserves the algebraic regime to which the corresponding spine element belongs.

**Proposition 7.1** (Certified Metamorphic Invariance). *Let*

$$m \geq 2,$$

*and suppose that*

$$q \equiv q_m^* \pmod{2^{\kappa_m}}.$$

*For every integer*

$$t \geq 0,$$

*define*

$$q_t = q + 2^{\kappa_m} t$$

and

$$n_t = 2^m q_t - 1.$$

Then

$$\tau(n_t) = m.$$

*Proof.* Since

$$q_t = q + 2^{\kappa_m} t,$$

one has

$$q_t \equiv q \pmod{2^{\kappa_m}}.$$

Because

$$q \equiv q_m^* \pmod{2^{\kappa_m}},$$

it follows that

$$q_t \equiv q_m^* \pmod{2^{\kappa_m}}.$$

Thus

$$2^m q_t - 1 \in \mathcal{D}_m.$$

By Definition 4.3,

$$\tau(n_t) = m.$$

□

The same transformation preserves shifted non-certified classes. In that case it does not preserve exact descent at the first exit step; instead, it preserves the first-exit valuation determined by the shift.

**Proposition 7.2** (Shifted Metamorphic Valuation Invariance). *Let*

$$m \geq 2$$

and let

$$1 \leq a < 2^{\kappa_m}.$$

Suppose that

$$q \equiv q_m^* + a \pmod{2^{\kappa_m}}.$$

For every integer

$$t \geq 0,$$

define

$$q_t = q + 2^{\kappa_m} t.$$

Then

$$v_2(3^m q_t - 1) = v_2(a).$$

*Proof.* Since

$$q_t = q + 2^{\kappa_m} t,$$

one has

$$q_t \equiv q \pmod{2^{\kappa_m}}.$$

Therefore

$$q_t \equiv q_m^* + a \pmod{2^{\kappa_m}}.$$

By Definition 5.4,

$$v_2(3^m q_t - 1) = v_2(a).$$

□



Thus the same residue-preserving transformation has two different effects depending on the class. On the certified class

$$\mathcal{D}_m,$$

it preserves exact first descent:

$$\tau(2^m q_t - 1) = m.$$

On each shifted non-certified class

$$\mathcal{R}_{m,a}, \quad 1 \leq a < 2^{\kappa_m},$$

it preserves the first-exit valuation:

$$v_2(3^m q_t - 1) = v_2(a).$$

Consequently, the certified and non-certified regimes are parallel at the modular level but differ at the first possible exit step: the certified class satisfies the valuation threshold and descends at step  $m$ , whereas each shifted non-certified class has valuation

$$v_2(a) < \kappa_m$$

and therefore fails the certified first-exit mechanism.

This metamorphic structure is used in the computational layer only as a post-evaluation framework. It does not replace exact integer iteration and is not used to prove descent. Its role is to organize comparisons between residue-preserving transformations of certified cases and corresponding shifted non-certified controls.

## 7.4 AAS-Based Metamorphic Post-Evaluation

An AAS-style metamorphic post-evaluation was computed after the exact-integer descent analysis. The AAS transformation used here is adapted from the Artificial Age Score framework [9]. Metamorphic relations are used as consistency principles in software testing and validation, where transformed inputs are expected to preserve or predict corresponding output relations [7, 8]. In the present setting, the AAS layer is used only as a diagnostic post-evaluation tool. It is not used as proof of descent.

For each tested spine-level case

$$n = 2^m q - 1,$$

five deviation magnitudes were computed. These quantities measure deviations from the certified exact-descent structure.

The certified-residue deviation is the normalized cyclic distance from the certified residue class:

$$\Delta_{\text{res}}(q) = \frac{\min\{d, 2^{\kappa_m} - d\}}{2^{\kappa_m} - 1},$$

where

$$d \equiv q - q_m^* \pmod{2^{\kappa_m}}, \quad 0 \leq d < 2^{\kappa_m}.$$

The valuation-threshold deviation is

$$\Delta_{\text{val}}(q) = \frac{\max\{0, \kappa_m - v_2(3^m q - 1)\}}{\kappa_m}.$$

This quantity is zero precisely when the certified valuation threshold is met.

The pre-descent deviation is

$$\Delta_{\text{pre}}(n) = \begin{cases} 0, & T^j(n) > n \text{ for all } 1 \leq j \leq m-1, \\ \max_{1 \leq j \leq m-1} \frac{\max\{0, n - T^j(n)\}}{n}, & \text{otherwise.} \end{cases}$$

For spine elements, the algebraic persistence identity implies

$$\Delta_{\text{pre}}(n) = 0$$

throughout the tested range.

The exit-descent deviation is

$$\Delta_{\text{exit}}(n) = \frac{\max\{0, T^m(n) - n\}}{n}.$$

This quantity is zero exactly when the first possible exit iterate satisfies

$$T^m(n) \leq n.$$

In the certified setting, the stronger inequality

$$T^m(n) < n$$

holds by the valuation-threshold theorem.

The exact-first-descent deviation is

$$\Delta_{\tau}(n) = \begin{cases} 1, & \tau(n) \text{ is unresolved within the iteration limit,} \\ 0, & \tau(n) = m, \\ \min\{1, \frac{|\tau(n) - m|}{m}\}, & \tau(n) \text{ is resolved and } \tau(n) \neq m. \end{cases}$$

Thus  $\Delta_{\tau}$  measures deviation from exact first descent at the first possible exit step.

Each deviation magnitude

$$\Delta_i$$

was transformed into a bounded consistency value

$$x_i = \frac{1}{1 + \Delta_i}.$$

The logarithmic penalty kernel was then applied:

$$\phi(x_i) = -\log_2 \left( \frac{x_i + \varepsilon}{1 + \varepsilon} \right), \quad \varepsilon = 10^{-6}.$$

Using equal weights and redundancy-neutral scoring,

$$R_i = 0,$$

the AAS value was computed as

$$AAS(n) = \frac{1}{5} \sum_{i=1}^5 \phi(x_i),$$

where the five inputs correspond to

$$\Delta_{\text{res}}, \quad \Delta_{\text{val}}, \quad \Delta_{\text{pre}}, \quad \Delta_{\text{exit}}, \quad \Delta_{\tau}.$$

Under this definition,

$$AAS(n) = 0$$

if and only if all five deviation magnitudes are zero. In particular, every certified case satisfies

$$\Delta_{\text{res}} = \Delta_{\text{val}} = \Delta_{\text{pre}} = \Delta_{\text{exit}} = \Delta_{\tau} = 0,$$

and therefore has

$$AAS(n) = 0.$$

For tested non-certified spine cases, the residue and valuation-threshold deviations are positive. Delayed descent also contributes through

$$\Delta_{\tau}$$

when

$$\tau(n) > m.$$

Thus the AAS value provides a severity-sensitive diagnostic score separating certified exact descent from non-certified delayed descent in the finite computation. This separation is empirical and finite-range only; it is not an infinite descent theorem.

## 7.5 Control Groups and AAS Results

The AAS post-evaluation was applied to four groups. Two of these were the principal groups in the finite partition:

$$\mathcal{D}_m(10^7)$$

and

$$\mathcal{N}_m(10^7).$$

The certified group consists of spine parameters satisfying

$$q \equiv q_m^* \pmod{2^{\kappa_m}},$$

whereas the full non-certified group consists of parameters satisfying

$$q \not\equiv q_m^* \pmod{2^{\kappa_m}}.$$

Together, these two groups form the non-overlapping finite partition of the tested spine-level population.

Two auxiliary diagnostic controls were also evaluated. The boundary-shifted control was defined by

$$q \equiv q_m^* + 1 \pmod{2^{\kappa_m}}.$$

This is the nearest shifted non-certified residue class to the certified class and satisfies

$$v_2(3^m q - 1) = v_2(1) = 0$$

by Definition 5.4. The random-spine control consisted of sampled non-certified spine parameters in the same search range, with certified parameters excluded. These two controls are subsets or samples within the non-certified regime; they are not additional disjoint components of the main partition.

The finite computation up to

$$N = 10^7$$

produced the summary shown in Table 4.

The certified group produced

$$AAS = 0$$

| Group              | Cases     | Unresolved | Exact at $m$ | Delayed   | Mean $\tau$ | Mean $E$ | Max $E$ | Mean AAS |
|--------------------|-----------|------------|--------------|-----------|-------------|----------|---------|----------|
| Certified          | 1,068,631 | 0          | 1,068,631    | 0         | 2.6068      | 0.0000   | 0       | 0.0000   |
| Boundary-shifted   | 1,068,630 | 0          | 0            | 1,068,630 | 8.9187      | 6.3120   | 152     | 0.7871   |
| Random spine       | 3,101     | 0          | 0            | 3,101     | 24.0568     | 14.7372  | 127     | 1.3866   |
| Full non-certified | 3,931,361 | 0          | 0            | 3,931,361 | 9.8429      | 6.7360   | 153     | 0.7587   |

Table 4: Finite-range delayed-descent and AAS summary up to  $N = 10^7$ . The certified and full non-certified rows form the principal finite partition. Boundary-shifted and random-spine rows are auxiliary diagnostic controls in the non-certified regime.

in every tested case. It also had no unresolved cases, no delayed cases, and exact first descent at step  $m$  throughout the tested range. In contrast, all evaluated non-certified groups had positive mean AAS, no exact descents at step  $m$ , and only delayed descents among resolved cases.

The full non-certified complement contained

$$3,931,361$$

tested cases. All were resolved within the iteration limit, none descended at the first possible exit step, and all exhibited delayed descent. The auxiliary controls showed the same qualitative separation from the certified regime: boundary-shifted cases were all delayed, and random-spine cases were all delayed among the sampled non-certified parameters.

The corresponding mean deviation components are shown in Table 5.

| Group              | $\overline{\Delta_{\text{val}}}$ | $\overline{\Delta_{\text{exit}}}$ | $\overline{\Delta_{\tau}}$ |
|--------------------|----------------------------------|-----------------------------------|----------------------------|
| Certified          | 0.000000                         | 0.000000                          | 0.000000                   |
| Boundary-shifted   | 1.000000                         | 2.148121                          | 0.830660                   |
| Random spine       | 0.851261                         | 128.673983                        | 0.817874                   |
| Full non-certified | 0.827930                         | 2.684171                          | 0.815357                   |

Table 5: Mean valuation, exit, and exact- $\tau$  deviation components for certified and non-certified diagnostic groups.

The certified group has zero valuation, exit, and exact- $\tau$  deviation throughout the tested range. The boundary-shifted group has

$$\overline{\Delta_{\text{val}}} = 1,$$

which reflects complete failure of the certified valuation threshold in the shift

$$a = 1.$$

The random-spine group has a larger mean exit deviation, reflecting the greater variability of generic non-certified spine behavior. The full non-certified complement has positive mean deviation in all displayed diagnostic components while remaining fully resolved within the tested range.

These results are consistent with the algebraic structure developed above. Certified elements satisfy the valuation threshold and descend at the first possible exit step. Shifted and generic non-certified elements fail the certified first-exit threshold; within the finite range

$$N = 10^7,$$

they exhibit delayed descent rather than exact first-exit descent.

The AAS layer is diagnostic rather than deductive. It is a severity-sensitive post-evaluation of finite exact-integer computations. It is not a proof of universal descent and not a proof that all non-certified spine elements eventually descend.

## 7.6 Interpretation of the AAS Separation

The AAS post-evaluation provides a diagnostic summary of the separation between certified exact descent and non-certified delayed descent in the tested range. Its interpretation follows the algebraic decomposition

$$S_m = \mathcal{D}_m \sqcup \mathcal{N}_m.$$

On the certified class

$$\mathcal{D}_m,$$

the residue condition implies the valuation threshold

$$v_2(3^m q - 1) \geq \kappa_m.$$

By Definition 4.1, this gives

$$\tau(2^m q - 1) = m.$$

Therefore the residue, valuation, pre-descent, exit, and exact- $\tau$  deviations all vanish. Consequently,

$$\text{AAS}(2^m q - 1) = 0$$

for every tested certified case.

The shifted non-certified classes exhibit a different first-exit structure. If

$$q \equiv q_m^* + a \pmod{2^{\kappa_m}}, \quad 1 \leq a < 2^{\kappa_m},$$

then Definition 5.4 gives

$$v_2(3^m q - 1) = v_2(a).$$

Since

$$v_2(a) < \kappa_m,$$

the certified valuation threshold fails on every shifted non-certified class. Thus the non-certified classes are separated from the certified class already at the first-exit valuation level.

The boundary-shifted control corresponds to

$$a = 1.$$

Hence

$$v_2(3^m q - 1) = 0$$

throughout this class. This explains the observed positive AAS values in the boundary-shifted group: although the class is adjacent to the certified residue in modular space, it has the maximal valuation-threshold deviation.

The full non-certified complement showed the same qualitative separation in the finite computation up to

$$N = 10^7.$$

All tested full non-certified cases had positive AAS, resolved first-descent times, and delayed descent

$$\tau(n) > m.$$

Thus the finite computation exhibited the diagnostic pattern

$$AAS_{\text{certified}} = 0, \quad AAS_{\text{noncertified}} > 0,$$

for certified, boundary-shifted, random-spine, and full non-certified comparisons.

This separation should be interpreted carefully. For the chosen AAS deviations, the implication

$$n \in \mathcal{D}_m \implies AAS(n) = 0$$

holds structurally; in the computation it was verified across the tested finite range. The positive AAS values in the non-certified groups indicate failure of the certified first-exit pattern and quantify the severity of that failure in the tested range. They do not constitute a proof that all non-certified elements descend, nor do they imply global convergence of the Collatz map.

The role of AAS is therefore complementary to the algebraic results. The valuation-threshold theorem proves exact descent on

$$\mathcal{D}_m.$$

The shift valuation lemma describes the first-exit valuation on

$$\mathcal{N}_m.$$

The AAS layer summarizes, in a severity-sensitive way, how these two regimes separate under finite exact-integer evaluation.

## 8 Discussion and Limitations

The results of this paper are local rather than global. They establish exact first-descent certificates inside modular resistance spines for the accelerated odd Collatz map and describe the first-exit valuation structure of the complementary non-certified residue classes. The main algebraic contribution is the conversion of pre-descent resistance into an exact first-descent certificate through the finite threshold

$$\kappa_m = \left\lceil \log_2 \left( \frac{3^m}{2^m - 1} \right) \right\rceil.$$

For spine elements

$$n = 2^m q - 1,$$

the persistence identity forces

$$\tau(n) \geq m$$

whenever the first-descent time exists. The valuation threshold then identifies an explicit residue class of parameters  $q$  for which this lower bound is sharp. In this certified class,

$$q \equiv q_m^* \pmod{2^{\kappa_m}}, \quad q_m^* \equiv (3^m)^{-1} \pmod{2^{\kappa_m}},$$

one has

$$\tau(2^m q - 1) = m.$$

Thus each spine contains an infinite arithmetic subclass whose first-descent time is exactly the first possible exit step.

The complementary class

$$\mathcal{N}_m = S_m \setminus \mathcal{D}_m$$

is not certified by this threshold. The shifted valuation lemma shows that its first-exit valuation is nevertheless structured. If

$$q \equiv q_m^* + a \pmod{2^{k_m}}, \quad 1 \leq a < 2^{k_m},$$

then

$$v_2(3^m q - 1) = v_2(a).$$

Hence the failure of the certified threshold in the non-certified complement is algebraic rather than random. However, this result concerns only the first possible exit step. It does not determine the later accelerated odd iterates and therefore does not prove descent for all elements of

$$\mathcal{N}_m.$$

This distinction is essential. The certified class

$$\mathcal{D}_m$$

is handled by an infinite algebraic theorem. The full non-certified complement

$$\mathcal{N}_m$$

is handled in this paper only by finite exact-integer computation. Up to

$$N = 10^7,$$

all tested non-certified elements were resolved, none descended at the first possible exit step, and all exhibited delayed descent. This provides finite-range evidence for delayed descent in the complement, but it is not an infinite theorem.

The AAS-based metamorphic layer has the same limitation. It provides a severity-sensitive diagnostic separation between certified and non-certified behavior in the tested range. Certified cases have zero AAS because the residue, valuation, pre-descent, exit, and exact-descent conditions all agree with the algebraic certificate. Non-certified controls have positive AAS because they fail the certified first-exit pattern and, in the tested range, exhibit delayed descent. This diagnostic separation is useful for organizing the finite computation, but it is not a substitute for a proof of descent.

The main limitation is therefore the absence of an infinite delayed-descent theorem for

$$\mathcal{N}_m.$$

The shifted valuation lemma gives exact control of

$$v_2(3^m q - 1)$$

at the first exit step, but a proof of descent in the non-certified complement would require control of later valuations along the accelerated odd trajectory. In particular, one would need a mechanism showing that for every non-certified spine element there exists some

$$r > m$$

such that

$$T^r(n) < n.$$

The finite computation suggests this behavior up to the tested bound, but does not prove it beyond that range.

Consequently, the results should not be interpreted as a proof of the Collatz conjecture. They prove exact first descent for explicit infinite certified subclasses and give an algebraic description of the first-exit obstruction in the remaining classes. A remaining structural problem is to convert delayed-descent evidence for the full non-certified complement into an infinite coverage theorem within the spine framework.

## 8.1 Local Nature of the Certificate

The exact-descent certificate is local to a fixed modular resistance spine. It applies to integers of the form

$$n = 2^m q - 1$$

whose spine parameter satisfies the certified congruence

$$q \equiv q_m^* \pmod{2^{\kappa_m}}, \quad q_m^* \equiv (3^m)^{-1} \pmod{2^{\kappa_m}}.$$

For each fixed

$$m \geq 2,$$

this congruence defines the infinite arithmetic class

$$\mathcal{D}_m = \{ 2^m(q_m^* + 2^{\kappa_m}t) - 1 : t \geq 0 \}.$$

Every element of this class satisfies

$$\tau(n) = m.$$

The complement

$$\mathcal{N}_m = S_m \setminus \mathcal{D}_m$$

is also infinite. Its shifted residue classes have an exact first-exit valuation structure:

$$q \equiv q_m^* + a \pmod{2^{\kappa_m}}, \quad 1 \leq a < 2^{\kappa_m},$$

which implies

$$v_2(3^m q - 1) = v_2(a).$$

This is an infinite algebraic statement about the first possible exit step, not an infinite descent theorem for the non-certified complement.

Thus the paper proves two infinite local results: exact first descent on

$$\mathcal{D}_m,$$

and exact first-exit valuation structure on the shifted residue decomposition of

$$\mathcal{N}_m.$$

The observed delayed descent of the full non-certified complement remains a finite-range exact-integer result. It does not establish that every element of

$$\mathcal{N}_m$$

eventually descends, and it does not imply global convergence of the Collatz map.

## 8.2 Relation to the Collatz Conjecture

The Collatz conjecture remains open. General background on the  $3x + 1$  problem is provided in [2, 3], and a modern logarithmic-density result for almost all Collatz orbits is given in [5]. The results of the present paper do not establish global convergence.

For the accelerated odd map, a universal first-descent theorem would imply the Collatz conjecture by the standard minimal-counterexample argument. Namely, if every positive odd integer

$$n > 1$$



satisfied

$$\exists r \geq 1 : T^r(n) < n,$$

then no minimal odd counterexample to the Collatz conjecture could exist.

The present paper proves a more restricted local statement. For spine elements

$$n = 2^m q - 1,$$

it establishes the implication

$$v_2(3^m q - 1) \geq \kappa_m \implies \tau(2^m q - 1) = m.$$

Equivalently, it proves exact first descent on the certified residue class

$$q \equiv q_m^* \pmod{2^{\kappa_m}}.$$

This gives infinitely many exact first-descent examples in every modular resistance spine, but it does not cover all odd integers.

The non-certified complement is also not resolved by an infinite theorem in this paper. Although the shifted valuation lemma proves that

$$q \equiv q_m^* + a \pmod{2^{\kappa_m}}, \quad 1 \leq a < 2^{\kappa_m},$$

implies

$$v_2(3^m q - 1) = v_2(a),$$

this describes only the first possible exit valuation. It does not prove that every non-certified element eventually descends.

The finite computation up to

$$N = 10^7$$

showed that every tested case in the full non-certified complement had a resolved first-descent time and exhibited delayed descent. This provides finite-range exact-integer evidence, but it does not prove the universal delayed-descent statement

$$\forall q \geq 1, \quad q \not\equiv q_m^* \pmod{2^{\kappa_m}} \implies \exists r > m : T^r(2^m q - 1) < 2^m q - 1.$$

Nor does it prove

$$\forall n \in \mathbb{O}^+, n > 1, \quad \exists r \geq 1 : T^r(n) < n.$$

Thus the paper should be interpreted as establishing a local certified-descent framework. It proves infinite exact-descent subclasses, proves an infinite first-exit valuation structure for shifted non-certified classes, and reports finite-range delayed-descent evidence for the full non-certified complement. It does not prove that every orbit enters a certified subclass, and it does not prove the Collatz conjecture.

### 8.3 Role of Computation

The exact-integer computation up to

$$N = 10^7$$

has three roles: verification of the finite certified population, exploration of the finite non-certified complement, and diagnostic comparison through AAS-based metamorphic post-evaluation. The numerical totals and delay statistics are reported in Tables 1, 3 and 4.

First, the computation verifies the certified exact-descent subclasses within the tested range. Every certified case was checked by exact iteration and satisfied

$$\tau(n) = m.$$

This agrees with the algebraic prediction of Definition 4.1. The finite verification is therefore an implementation check and a finite-range confirmation, not a substitute for the theorem.

Second, the computation examines the full non-certified complement. In the tested range, every non-certified case was resolved, none descended at the first possible exit step, and all exhibited delayed descent:

$$\tau(n) > m.$$

The delayed-descent excess

$$E(n) = \tau(n) - m$$

was used to quantify the amount of delay beyond the first possible exit step. These data provide finite-range evidence for delayed descent in

$$\mathcal{N}_m,$$

but they do not prove that all elements of

$$\mathcal{N}_m$$

eventually descend.

Third, the computation supports an AAS-based diagnostic comparison among certified, boundary-shifted, random-spine, and full non-certified groups. The certified group produced zero AAS throughout the tested range, while the non-certified diagnostic groups produced positive AAS values. This separation reflects failure of the certified valuation threshold and delayed first descent in the tested non-certified cases.

The algebraic and computational roles are therefore distinct. The exact-descent theorem is proved by combining spine persistence with the valuation-threshold inequality. The shifted valuation lemma is also algebraic. The computation verifies implementation consistency, records the finite distribution of certified and non-certified cases, and provides finite-range evidence for delayed descent in the full non-certified complement. This use of exact finite verification is consistent with computational approaches in the  $3x + 1$  literature [4].

## 8.4 Role of AAS

The AAS-based metamorphic post-evaluation is used as a diagnostic layer. It is not part of the proof of exact descent and is not used to infer universal descent. Its purpose is to quantify, in a severity-sensitive way, how far a tested case deviates from the certified exact-descent pattern. This use of transformed input classes as consistency checks is related to the general methodology of metamorphic testing [7, 8].

In the certified regime, the residue condition

$$q \equiv q_m^* \pmod{2^{\kappa_m}}$$

implies the valuation threshold and hence

$$\tau(2^m q - 1) = m.$$

Consequently, all AAS deviation components vanish:

$$\Delta_{\text{res}} = \Delta_{\text{val}} = \Delta_{\text{pre}} = \Delta_{\text{exit}} = \Delta_{\tau} = 0,$$

so every tested certified case has

$$\text{AAS} = 0.$$

In the non-certified diagnostic groups, the certified residue condition fails, and the valuation-threshold deviation is positive. In the tested range, boundary-shifted, random-spine, and full non-certified groups also exhibited delayed descent. These deviations produce positive AAS values and yield a finite-range diagnostic separation between certified exact descent and non-certified delayed descent.

Thus AAS functions as a post-evaluation and consistency-audit framework. It summarizes the computational separation already observed by exact integer iteration, but it does not replace the algebraic proof of the certified theorem, and it does not prove that all non-certified elements eventually descend.

## 8.5 Sparsity of Certified Subclasses

For each fixed

$$m \geq 2,$$

the certified exact-descent class occupies one residue class modulo

$$2^{\kappa_m}$$

among the spine parameters  $q$ . Its relative density inside the  $m$ -th spine is therefore

$$2^{-\kappa_m}.$$

Since

$$\kappa_m = m \log_2 \left( \frac{3}{2} \right) + O(1),$$

this density has the asymptotic scale

$$2^{-\kappa_m} \asymp \left( \frac{2}{3} \right)^m.$$

Thus certified exact-descent subclasses become sparse as the spine level  $m$  increases.

This sparsity reflects the strength of the certificate. The theorem does not merely prove eventual descent; it proves descent at the first possible exit step:

$$\tau(2^m q - 1) = m.$$

Such exact first-exit descent requires a precise 2-adic alignment,

$$v_2(3^m q - 1) \geq \kappa_m,$$

and therefore selects only one residue class modulo

$$2^{\kappa_m}.$$

The full non-certified complement has complementary relative density

$$1 - 2^{-\kappa_m}.$$

Hence most spine parameters are not certified by the first-exit threshold. This does not contradict the theorem and does not indicate non-descent. It only shows that most spine elements require analysis beyond the first possible exit step.

The shifted valuation structure organizes this complement:

$$q \equiv q_m^* + a \pmod{2^{\kappa_m}}, \quad 1 \leq a < 2^{\kappa_m},$$

with

$$v_2(3^m q - 1) = v_2(a).$$

Thus the sparse certified class and the large non-certified complement are parts of the same modular structure. The certified class gives guaranteed exact first-exit descent, while the non-certified classes provide the structured setting for delayed-descent analysis.

## 8.6 Future Directions

The present paper certifies exact first descent at the first possible exit step for one residue class inside each modular resistance spine. A natural next step is to develop multi-stage or cumulative valuation certificates that apply after the first exit step.

For an accelerated odd trajectory

$$n_0 = n, \quad n_{i+1} = T(n_i),$$

define

$$a_i = v_2(3n_i + 1)$$

and the cumulative valuation

$$A_r(n) = \sum_{i=0}^{r-1} a_i.$$

A descent criterion over  $r$  accelerated odd steps should compare the accumulated division by powers of 2 with the accumulated multiplication by powers of 3. In an exact treatment, this comparison must also account for the additive terms generated by the repeated +1 in the Collatz map.

One possible formulation is to seek explicit conditions on

$$A_r(n)$$

and the intermediate accumulated valuations that imply

$$T^r(n) < n.$$

Such a criterion would extend the single-exit threshold

$$v_2(3^m q - 1) \geq \kappa_m$$

to a multi-step setting. This would be especially relevant for non-certified spine elements, where the first-exit valuation satisfies

$$v_2(3^m q - 1) < \kappa_m$$

but later accelerated odd iterates may supply enough additional 2-adic division to force descent.

The most immediate open problem is therefore an infinite delayed-descent theorem for the full non-certified complement:

$$\forall m \geq 2, \quad \forall n \in \mathcal{N}_m, \quad \exists r > m : T^r(n) < n.$$

The finite computation up to

$$N = 10^7$$

supports this behavior in the tested range, but no such theorem is proved here.

A broader direction is a first-descent coverage theorem for all positive odd integers:

$$\forall n \in \mathbb{O}^+, n > 1, \quad \exists r \geq 1 : T^r(n) < n.$$

Such a theorem would imply the Collatz conjecture by the minimal-counterexample argument. The results of the present paper do not prove this statement. They provide local exact-descent certificates and a structured description of the first-exit obstruction that any future coverage theorem would need to overcome.

Another possible direction is to study entry into certified subclasses. One may ask whether arbitrary accelerated odd trajectories eventually enter some certified class, in the sense that for some iterate  $k \geq 0$  and some spine level  $m \geq 2$ ,

$$T^k(n) \in \mathcal{D}_m.$$

This is stronger than the local results proved here and is not established in the present paper. It remains a separate open problem from the finite-range delayed-descent evidence reported above.

Thus the main future objective is to pass from first-exit certification to delayed-descent coverage: first for the full non-certified complement of each spine, and ultimately for all accelerated odd trajectories.

## 9 Conclusion

This paper established local exact first-descent certificates inside modular resistance spines for the accelerated odd Collatz map. For spine elements

$$n = 2^m q - 1, \quad m \geq 2,$$

the persistence identity excludes descent before the  $m$ -th accelerated odd iterate. The exit formula then yields an explicit valuation threshold

$$\kappa_m = \left\lceil \log_2 \left( \frac{3^m}{2^m - 1} \right) \right\rceil$$

such that

$$v_2(3^m q - 1) \geq \kappa_m \implies \tau(2^m q - 1) = m.$$

Thus the threshold converts pre-descent resistance into an exact first-descent certificate.

Equivalently, the certified congruence

$$q \equiv q_m^* \pmod{2^{\kappa_m}}, \quad q_m^* \equiv (3^m)^{-1} \pmod{2^{\kappa_m}},$$

defines an infinite exact-descent subclass

$$\mathcal{D}_m$$

inside every spine

$$S_m = \{n \in \mathbb{O}^+ : n \equiv -1 \pmod{2^m}\}.$$

Every element of this certified class satisfies

$$\tau(n) = m.$$

The full non-certified complement

$$\mathcal{N}_m = S_m \setminus \mathcal{D}_m$$

was also given an algebraic first-exit description. For shifted non-certified classes

$$q \equiv q_m^* + a \pmod{2^{\kappa_m}}, \quad 1 \leq a < 2^{\kappa_m},$$

the first-exit valuation is exactly

$$v_2(3^m q - 1) = v_2(a).$$

Hence the failure of the certified threshold in the non-certified complement is structured by the 2-adic valuation of the shift from the certified residue.

Exact-integer computation up to

$$N = 10^7$$

confirmed the certified theorem throughout the tested range and examined the full non-certified complement. The computation produced

$$1,068,631$$

certified cases, all with exact first descent at step  $m$ , and

$$3,931,361$$

full non-certified cases, all resolved and all exhibiting delayed descent rather than first-exit descent. The delayed-descent excess on the full non-certified complement satisfied

$$\bar{E} = 6.7360, \quad \text{median}(E) = 4, \quad \max E = 153.$$

An AAS-based metamorphic post-evaluation separated the same regimes diagnostically. Certified cases formed zero-AAS consistency classes, whereas boundary-shifted, random-spine, and full non-certified controls produced positive severity-sensitive scores. This AAS layer was used only as finite-range post-evaluation and not as a proof mechanism.

The results do not prove the Collatz conjecture. They prove infinite local exact-descent certificates, establish an infinite first-exit valuation structure for shifted non-certified classes, and provide finite-range exact-integer evidence for delayed descent in the full non-certified complement up to

$$N = 10^7.$$

The remaining challenge is to replace this finite delayed-descent evidence by an infinite coverage theorem for the non-certified complement, and ultimately for all accelerated odd trajectories.

## Code Availability

The Python implementation used for the exact-integer verification and AAS-based metamorphic post-evaluation is available in the author's public GitHub repository:

<https://github.com/seymus1/collatz-spine-exact-descent>

The script reproduces the summary statistics and tables reported in the manuscript.

## Author Contributions

Seyma Yaman Kayadibi is the sole author of this work and was responsible for the conceptualization, mathematical formulation, computational implementation, analysis, interpretation of results, and manuscript preparation.

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